Journal of Computer Information Systems

ISSN: (Print) (Online) Journal homepage: https://www.tandfonline.com/loi/ucis20

Machine Learning Based Product Classification for eCommerce

Mieczysław Pawłowski

JOURNAL OF COMPUTER INFORMATION SYSTEMS
https://doi.org/10.1080/08874417.2021.1910880

Taylor & Francis
Taylor & Francis Group

Machine Learning Based Product Classification for eCommerce

Mieczysław Pawłowski

Maria Curie–Sklodowska University, Lublin, Poland

ABSTRACT
Product research in web shops involves categories as a helper system for search and browsing. The categorized offer reduces user's mistakes and misled presentation. Hence, learning a product classifier to work with high precision and proper recall is of fundamental importance in order to provide high-quality shopping experience. However, products can be perfectly described they can miss users' vocabulary when they do research. Thus, users' content in form of search phrases has been utilized in order to extent a product data. Thus, original product names and classes have been supplemented with users' search phrases, which changed classification accuracy in significantly positive direction but also slightly influencing on data consistency. The article presents new knowledge, research framework, machine learning classifiers selection and result analysis with implication for academia and some suggestions for business practitioners.

KEYWORDS
Machine learning; product data; product classification; e-commerce

Introduction

Nowadays, consumers and businesses make use of e-commerce continuously as a natural tool for market research and purchase. A search for products and catalog browsing is the core behavior of the purchase process which users do in web shops. Products are identified by their forms, functions, and behaviors.[1] They are expressed in form of textual product name and description, digital media content, and classification, which play an important role in decision-making during customer journey. All these factors strongly influence customer convenience, technology acceptance, and finally customer experience. Recently, not only the company but also customers are willing to deliver a product-related user-generated content (UGC) in form of scoring notes, opinions review, and the intention to help others in evaluation of product functionalities.[2] UGC refers to the content generated from regular users who voluntarily contribute data, information, or media to social media platforms where this content is then displayed in a useful or entertaining way.[3] Massive amount of UGC has a profound influence on the business world and society. Today's customers in the business world are much more likely to consider product or brand-related UGC produced by other customers rather than rely on marketing material. Thus, trust among customers can be leveraged, and tracking customers' willingly shared product- or brand-related UGC has the potential to provide gains over contemporary customer communications. UGC has also decentralized nature due to the way people produce and consume content.[4] In this study, we utilize search phrases as the form of UGC specific for product search in the web shops. Scientific investigations related to the activity of web shops' users are continuously reported,[5–7] which confirms that study of the role of UGC which they deliver in different forms can help in recognizing customers' latent expectations encapsulated in search phrase, which was used in memory or entered to the system when they are searching and browsing product offer in a web shop.

Presented paper fill some gaps in literature concerning product classification and product data consistency maintenance for web shops with application of machine learning classifiers (MLC) and UGC as a moderating factor of product classification metrics. Thus, we analyze product names and classes as a short textual object which can be successfully filtered, mined, and classified using MLC techniques.[8–10]

The rest of this article is organized as follows. In next chapter, we present literature review and hypothesis development. In section 3 the methodology, research model, metrics, and data sample are explained. In chapter 4, we describe numerical results and its interpretation. In conclusion section we end with some remarks, limitation, and future directions.

CONTACT Mieczysław Pawłowski m.pawlowski@umcs.pl Maria Curie-Sklodowska University, Pl. Marii Curie-Sklodowskiej 5, Lublin, Poland.

Literature review and hypothesis development

Scientific papers of recent years report the successful use of MLC algorithms in classification of variety of text types like tweets, customer review, posts, or e-mails, medical notes or even poems and scientific papers.[11,12] Product names for web shops, or users' SP are usually textual data of short length that make them possible to be presented on the screen of laptop, smartphone, or printed on fiscal receipt. The shortness of information, that customers see and have to interpret in web shops is common that rises financial and operational risk for customer and the web shop operator. Shortness comes not only from shops but also from users. It can mislead themself during product search, catalog browsing and final selection. The consequences of mislead purchase could be very dissatisfying, what shown an example of booking.com.[13] The portal complains about the poorness of the query, that customers enter when searching portal's offer, while at the same time they are facing an extremely important decision when reserve several dozens of rooms for a company event. Users provoke misleading shop behavior because they provide incomplete queries only barely identify the basic characteristics of the product. Thus, web portals strive to precise uses queries by multi-level classification and product grouping for helping users search for meaning.[14] From user's perspective, the effectiveness of data retrieval means delivering the results that each user expects individually. Thus, the judge of search and classification quality includes subjective bias based on user's specific interest.[15,16] The judge of a search accuracy is also contextual because user has access to competitive content for comparison, what makes search result adequacy perceived dynamically while the progress of knowledge collection about the object of the interest is progressing.[17]

Product selection in web shop can be supported by recommender systems, which provide prompt of options, product variants or sets.[18] If recommender system works beyond up-selling it can reduce customer effort, however, it rises challenges that can be also applied to product classification mechanisms: a) the needs to distinguish between random associations of products viewed but not purchased from products deliberately purchased together, b) capture short-range language patterns, that is used in search, and c) master user vocabulary.[19] Thus, recommender can interactively display suggestions in a drop-down list as the searcher types a query phrase. The suggestions can be retrieved based on similar queries submitted by other users or previously prepared classification variants where some can include UGC.[20]

Search phrase variants demonstrate process how user adjust terms from personal vocabulary to the product names (descriptions) or product classification until intended of search meet results. Product classification is a part of the research fields known as ontology and taxonomy matching. In taxonomy matching, data are annotated for relationship (not for meaning). If you are going to find an object in certain category, a matching algorithm has to establish the meaning in external data or in context within the schema. In contrast, the ontologies logical systems of axioms work for data annotation according to functional meaning.[21] When classification is created in the ontology manner similar items are grouped according to the functionality. During a search process, user focuses on matching expectations with functionalities, however only felt by her/himself. That is why, we can state that search phrases links hidden variables on user site with products features represented by text objects (product names). The query expansion mechanisms can help searchers refine their queries by recommending additional or alternative search terms based on grouping functionalities. User-side automated assistance can improve searcher performance (as measured by the number of relevant documents found) by approximately 20%.[22] Thus, effectiveness product classification should provide at least two major enhancements: a) recall the right products on the screen in a way that the collection contains as many products as possible but all responding to the customer's request, and b) limit the match (precision), to the resulting subset of objects representing customer request with the minimal number of a missed objects.[18] Thus, we formulate a research hypothesis as follows:

H1: Supplementing product names as textual data that represent real object's features and functionalities with UGC as the additional vocabulary of product description can significantly improve classification accuracy.

Despite helping technology plays an important role in delivering relevant results during search and browsing, the quality of the product database, as the source of content cannot be indignant. The good practice for online stores is to present products categorized for major user profiles.[23,24] If we look at the classification criteria from the customer perspective, we found focus on maximizing customer satisfaction by data clarity, high precision on acceptable recall which pretend to be self-service effective.[25] The conjunctions of user content (reviews) with product features provide to understand the rationale for purchase decision and helpfulness of online reviews.[26,27] Search phrases, that customers type include hidden variable related to all contextual factors interacting in the moment of action. Therefore, despite these phrases can be very well converting to purchase in the certain circumstances, they can be useless or harmful in another situation. For example, when customer is looking for item to replace in emergency situation, she/he is usually preferring exactly the same product, the same producer, brand, and functionality because this reduces risk of wrong choice. When the same user is looking for the same product for a new construction (design) she/he is going to look around to check possible options. Thus, we state, that user content can positively influence product classification and then browsing and searchability according to user vocabulary in certain context but in another could be harmful. Thus, user content could work both positively for one user group and negatively for other when set up match or mismatch certain context, as shown in example above. As the users' vocabulary represented by UGC also represents a significant number of users and their context of action, thus general application of users' content as the universal extension of product content (names or descriptions) can provoke data ambiguity. Thus, the hypothesis H2 is formulated as follows:

H2: Supplementing the professionally classified textual objects with UGC provide to ambiguous classification.

In regular business purchases, where customers repeat selection and context of purchase, it can be beneficial individual vocabulary utilization. Specific vocabulary can be obtained not only from user dialect bat also from manufacturer's catalog. Thus, this manner can produce product catalog personalization as well as data ambiguity. Therefore, product managers and web shop operators would face a dilemma when UGC from one perspective adds value, but from other tilt product information to subjective side of certain customer group or even provide data worse quality.[28]

The positive influence of UGC on the search results can trigger the company professionals for systematic work for the product description and classification improvements in order to satisfying customers expectation, which the one is match their vocabulary used for search with search results. This will finally define successful self-service concept in the e-commerce business.[29]

Method

Overview

User activity starts when search phrase in typed to the search box. Web shop starts searching in product database. Users see results on a screen. When user decides to buy a one or several products from the search results, then the search phrase is market as supplementing product data. Product classification is utilized as helper system during search and for browsing, thus classification experiments reflect search process. Purchase process with remark on hypotheses H1 and H2 is presented in Figure 1.

Bag-of-words (BOW)[30] is specialized in mining of textual data as a tool of natural language processing including classification. The best results of text classification are reported in experiments when linear algorithms have been applied like multiclass logistic regression, decision trees, or support vector machine (SVM) with linear kernel.[11,31,32] Thus, we utilized BOW and term frequency (TF) and inverse document frequency IDF(t,D) as a weighting factors. TF(t,d) represents term t frequency in document d, what makes that TF value increases proportionally to the number of times where t appears in the document d. IDF(t,D) represents entropy of the corpus, which is calculates as

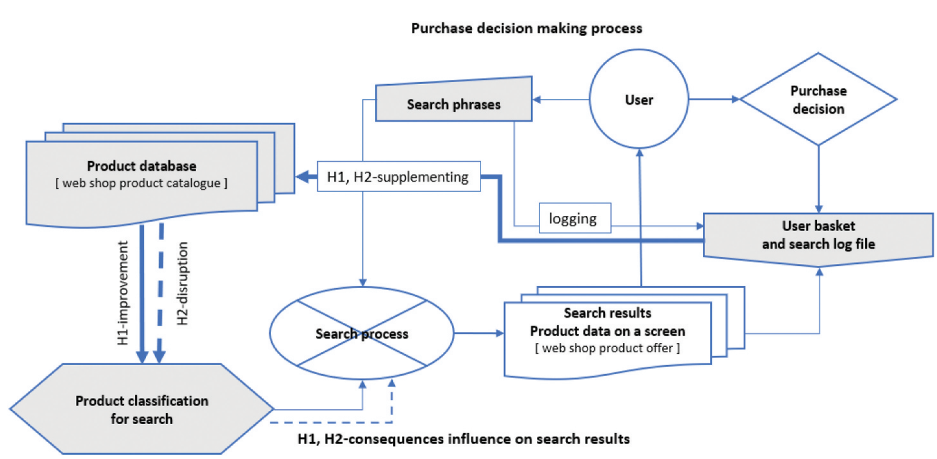

Figure 1. Product research process including search and purchase.

IDF(t,D) = log(N/(1+|{d in D: t in d}|)), where N is a total number of documents in the corpus N = |D|, and |{d in D: t in d}|). If term t appears many times with a corpus, IDF normalizes TF where the common term is weighting to be less meaningful then unique terms.[33]

Model

Data set, contain web shop's products (names & classes), which have been preprocessed, and after that split into training and test sub-sets and vectorizes in BOW models. After that classification algorithm have been trained and tested for product classification in data sub-sets (left part of the Figure 2). In the second phase of the research, we have used data set with UGC merged with for training and testing classification. Finally, we compared results from classification reports. Research model is presented in Figure 2.

Metrics

The quality of classification expresses the relationship between the result predicted by algorithms and classes annotated in real data, in our case by professional product managers. We have used accuracy, precision, recall, and F1-score for quality measurement. Accuracy relates true positives and true negatives to all products. True positives represent products that customer will purchase while true negatives represent products answering customer request but not purchased. If classification algorithm correctly recognizes true positives and true negatives, it could be described as working with good accuracy. Precision determines the ratio of true positives to the entire data brought to the user interface. True positives represent products that customer will purchase and products of the customer interest, but not purchased. False-positive occurs as the result of wrong selection. If we adjust system for high precision, we will downsize sub-set of proposals by eliminating random negatives. Over precise system will limit options for customer and downsize the effect of "looking around" for recognition. Recall represents the sub-set of true-positive in relation to the sum of true positive and false negatives. Higher recall should not only bring more products consistent with customer's interest, but also effectively filter false negatives. F1-score, balances precision and recall as a harmonic mean of both.[34]

Data sets

Data sets, in form of product file and a search log file as UGC, were obtained from the polish industry web shop. Product file was divided into five training data sets (see Table 1) and four test data sets (see Table 2). General characteristics of data sets are presented in Table 1 and 2.

A sample of several records of KwSem data set which is built based on Org and Wysz merged together is presented in Table 3. You can see Org sample and Wysz as well-represented UGC. It can be seen, that in some cases product name between items differs only with little

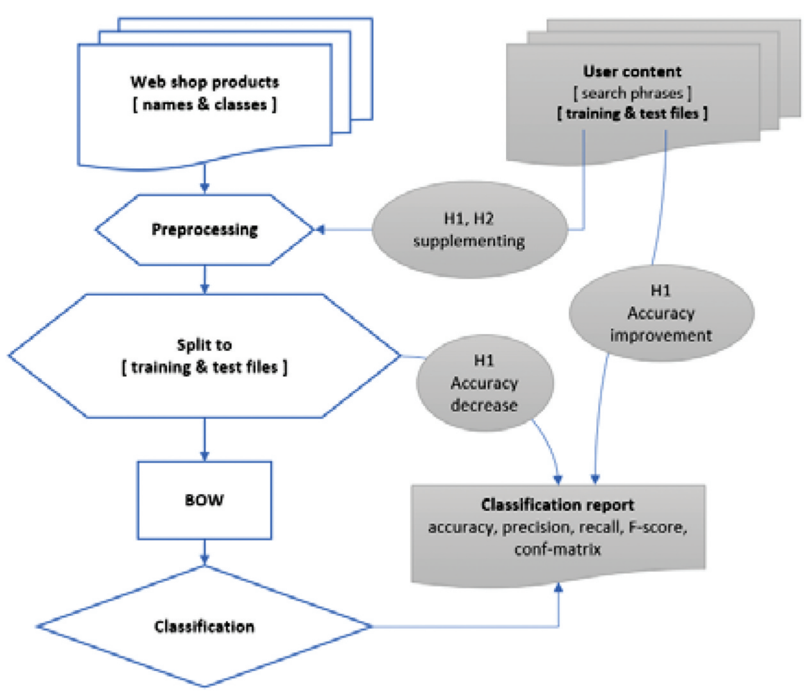

Figure 2. Research model.

Table 1. Training data sets, general characteristics.

Training file name	Training or test	Description	Number of items	Number of categories	Median length (char) item name/ number of words
Org	Training	Original web shop product register, without any preprocessing	28,179	512	50/7
Org2	Training	Preprocessing applied to file Org for filtering digits, special characters and then remove duplicates	10,399	467	34/5
Kw17	Training	Additional filtering applied to Org2 for removal categories with less than 17 records (25% outliers)	8,470	132	34/4
Kw17s	Training	Kw17 extended by adding, to the product description, unique phrases from Stempel stemmer	8,470	132	50/6
KwSem	Training	Org file merged with customer content (search phrases) (UGC)	65,271	512	24/4

Table 2. Test data sets, general characteristics.

Test file name	Training or test	Description	Number of items	Number of categories	Median length (char) item name/number of words
Kw4	Test	Test file – 25% of tail product register	7,374	383	44/6
Inv	Test	Test file – special offer items, no stock products, products on customer requests but compatible with main basket offer	19,535	132	36/5
Wysz	Test	Customer search phrases (UGC)	37,092	131	12/2
Wysz2	Test	Customer search phrases with preprocessing	10,280	127	12/3

Table 3. Org – product file and Wysz – user content in italic – sample from KwSem.

Data set	Class	Product description
Org	ADAPTER	AGT-16A 5-pin banana socket adapter
Org	ADAPTER	AGT-32A 5-pin banana socket adapter
Wysz	*ADAPTER*	*Basic banana adapter*
Wysz	*ADAPTER*	*Adapter switch construction switch*
Wysz	*ADAPTER*	*CIMA adapter*
Wysz	*ADAPTER*	*Danfoss AGT*
Org	ADAPTER	Adapter Cima modules white
Org	ADAPTER	Adapter Cima modules graphite connects
Org	ADAPTER	CIMA adapter for 2 screwless sockets K45 (45x45 mm), white
Org	ADAPTER	CIMA adapter for 2 K45 modules (45x45 mm) vertically, white
Org	INHIBITOR	Alpha −11 – protector inhibitor ant frost
Wysz	*INHIBITOR*	*Filter fluid with protector*
Org	INHIBITOR	Inhibitor corrosions do installation H.W. AGA-IK 1 L

Table 4. A descriptive statistic of user content (a phrases from the user's search log).

Phrase length (characters)	Count	1-word phrases	2-word phrases	3-word phrases	3+ words phrases
2	62	62	0	0	0
4	981	900	81	0	0
6	5 530	4 841	672	17	0
8	4 463	2 096	2 131	232	4
10	4 550	1 045	2 200	1 251	54
12	5 044	220	1 971	2 552	301
14	5 252	743	1 014	2 653	842
16	3 664	220	391	1 725	1 328
18	2 492	19	165	1 039	1 269
20	1 627	8	80	562	977
99	3 426	1	39	506	2 880
Total	37 091	10 155	8 744	10 537	7 655

details. We can also notice that UGC includes more general terms. Because they are search phrases, they are usually shorter than product names, what can be seen in last column of Tables 2 and 4. (median of the length (char) of item name = 12/number of words = 2–3). We believe they express the most important keywords for product search according to user's intent.

A descriptive statistic of user content is presented in Table 5. The length of the most users' search phrases is equal less than 16 characters (point of measure = quartile 3) and consists of three words or less (median equals 12 chars and 3 words) while in web shop median of the length of product name equals 50 characters and consist of seven words in average. That is why we hypothesis that the merge of such short content to product file could be harmful for classification based on this feature.

Classifiers

We have considered several classifiers; however, classifiers benchmarking is not the subject of this study. Based on Scikit learn[35] we tested 12 classifiers in performance test for the best accuracy, different algorithmic approach, and time efficiency. Our intention was to use three different methods of classification. Characteristics of tested algorithms are presented in Table 5.

We have selected classifiers for major investigation in procedure based on two data sets: Org – original product register and Kw17 – the most preprocessed. Test results are presented in Figure 3.

We can see time of calculation in sec in logarithmic scale on axis Time, and accuracy on axis Accuracy. Finally, Linear SVC, Ridge Classifier and Random Forest Classifier (in bold on Figure 3) has been selected for further experiments, because they obtained high accuracy, good efficiency in time, and they use different classification methods.

Table 5. Classifier's characteristics.

Classifier	Classifier description
Bernoulli NB	A Naive Bayes classifier for multivariate Bernoulli models. This classifier is suitable for discrete data. The difference is that while MultinomialNB works with occurrence counts, BernoulliNB is designed for binary/boolean features.
Complement NB	The Complement Naive Bayes classifier described in Rennie et al. (2003). The Complement Naive Bayes classifier was designed to correct the "severe assumptions" made by the standard Multinomial Naive Bayes classifier. It is particularly suited for imbalanced data sets.
K-Neighbors Class	Classifier implementing the k-nearest neighbor's vote.
Linear SVC	Linear Support Vector Classification. Similar to SVC with parameter kernel = 'linear', but implemented in terms of liblinear rather than libsvm, so it has more flexibility in the choice of penalties and loss functions and should scale better to large numbers of samples. This class supports both dense and sparse input, and the multiclass support is handled according to a one-vs-the-rest scheme.
Logistic Regression	Logistic regression is a linear model for classification, also known as logit regression, maximum-entropy classification, or the log-linear classifier. In this model, the probabilities describing the possible outcomes of a single trial are modeled using a logistic function.
Multinomial NB	The multinomial Naive Bayes classifier is suitable for classification with discrete features (e.g., word counts for text classification). The multinomial distribution normally requires integer feature counts. However, in practice, fractional counts such as tf-idf may also work.
Nearest Centroid	Each class is represented by its centroid, with test samples classified to the class with the nearest centroid.
Pass-Aggressive	The passive-aggressive algorithms are a family of algorithms for large-scale learning. They are similar to the Perceptron in that they do not require a learning rate. However, contrary to the Perceptron, they include a regularization parameter C.
Perceptron	The perceptron is classification algorithm suitable for large scale learning. By default, it does not require a learning rate, it is not regularized and updates its model only on mistakes.
Random Forest	A random forest is a meta estimator that fits a number of decision tree classifiers on various sub-samples of the data set and uses averaging to improve the predictive accuracy and control over-fitting.
Ridge Classifier	Classifier using Ridge regression. This classifier first converts the target values into {-1, 1} and then treats the problem as a regression task (multi-output regression in the multiclass case).
SGD Classifier	This estimator implements regularized linear models with stochastic gradient descent (SGD) learning: the gradient of the loss is estimated each sample at a time and the model is updated along the way with a decreasing strength schedule (aka learning rate). SGD allows minibatch (online/out-of-core) learning via the partial fit method.

Source: Scikit learn, Pedregosa et al.[35]

Results

Classification improvement through user content

For all results, a mean relative standard deviation (RSD) is equal 2,25% (quartile 1 = 0,25%, quartile 3 = 3,02%), thus all classification metrics were calculated as an average of those three classifiers. The classification accuracy is shown in Table 6. For training on data sets Org, Org2, Kw17, Kw17s, and testing on Org, Kw4, and Inv classification accuracy obtains value between 0,87 and 0,97 while for testing on Wysz and Wysz2 (search phrases) results decrease to the level of 0,37–0,49. This effect suggests that test subset of UGC do not match product names on the same level of accuracy as regular product catalog training subset does.

Testing Wysz and Wysz2 with classifiers trained on KwSem (Org + Wysz) shows classification accuracy on the same level (0,87–0,89) that was obtained in tests on regular product catalog subsets. This result suggests that UGC (Wysz file) merged to original product names (Org file) improve classification accuracy significantly (from 0.37–0.49 to 0.87–0.89). In Figure 4, it can be seen graphical demonstration of described relationships (left panel) and the differences in accuracy, precision, recall, and F1-score (right panel) to results obtained when classifiers where trained and tested on subsets taken from Org data set (the reference).

Classification experiments with training on data sets Org, Org2, Kw17, Kw17s and testing on Wysz,

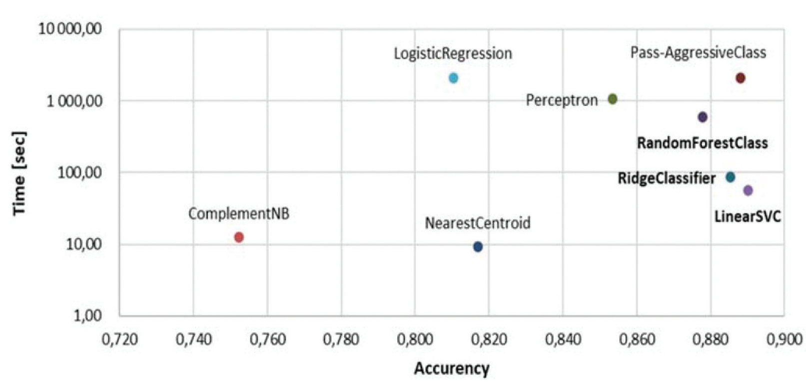

Figure 3. Classifier selection.

Table 6. Classification accuracy.

Training data set:	Org	Org2	Kw17	Kw17s	KwSem	
Test data set	Org	0.97	0.98	0.94	0.93	0.84
	Kw4	0.97				0.94
	Inv	0.87	0.89	0.89	0.88	0.87
	Wysz	0.47	0.49	0.49	0.48	0.87
	Wysz2	0.38	0.41	0.37	0.37	0.89

The measurement of precision, recall and F1-score is presented in Appendix.

Wysz2 shows 50% lower accuracy then model trained on KwSem. Training on KwSem shows even 5 to 10% increase of accuracy (Figure 4, right panel) to the results with training on Org-Kw17s data sets. The biggest drop occurs on accuracy, what means that relations of true positives (product that customer will buy) and true negatives (just looking at) to all data is worse than in reference experiments. The lowest drop occurs on precision, only 18,73%. It suggests that classifiers recognize users' phrases in product names (true positives), but not all. Recall is weaker because number of false negatives grew. Thus, we could summarize that classifiers do not sufficiently well recognize users' specific vocabulary until new knowledge have been delivered in Wysz file for training. This affects even the classification of catalog, but niche products. The classification accuracy when testing on Kw4, which is data set of tail products, increased from zero, where the models were trained of preprocessed data Org2, Kw17, Kw17s to 0,94 when classifiers were trained on KwSem. This can be interpreted as tail products names do not correlate with preprocessed files but correlate very well with user search phrases. Summarizing, we can state that hypothesis H1, which states that *supplementing product names as textual data that represent real object's features and functionalities with UGC as the additional vocabulary of product description can significantly improve classification accuracy*, is being verified positively.

Classification ambiguity

As presented in previous section, classifiers trained on KwSem data set to demonstrate high accuracy in product classification when testing on users' content (Wysz, Wysz2) or niche products Kw4. Despite that, we also faced the effect that the same model (trained on KwSem) shows decrease in accuracy when tested on Org data set (from 0,97 to 0,84), what can be seen in Figure 5, which shows accuracy of models trained on all train data sets and tested with Org file.

Thus, UGC may indicate ambiguity of training data. UGC if form of search phrases is very short, equal 1–2 words and 3–7 characters long. They may be annotated to different categories then the similar product groups in Org file. Classification in Org file is being made by professional product managers (PM). Although PM's work does not guarantee zero erroneous classification, we can surely state that UGC as search phrases inherit categories from products in the moment when they were added to the basket collecting a mental context the moments of search. For clarification, we have audited KwSem in order to find ambiguous records. There were detected 1,071 products where the category of the original items was different from the category matched with users' phrases. In Table 7 it is presented an example of similar products (similar product names) but assigned to different categories.

The phenomenon which we analyze here could be interpreted as the deterioration of a "clean" file caused by poor quality of added content due to its subjectivity and contextual nature, what we mentioned in literature review. A sample of erroneous categorization detected by classifiers is presented in Table 8 (confusion matrix).

Thus, rather a procedure of selective improvement of product data based on or with help of UGC should be applied then massive merge of UGC to product file. The consequence of merge of such a significant amount of

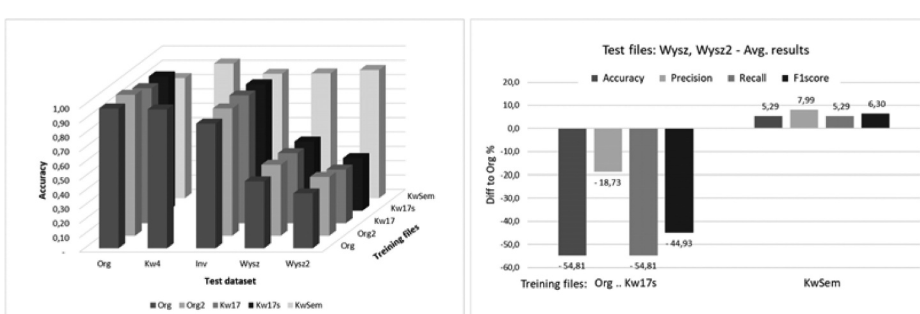

Figure 4. Classification accuracy in relation to data set used for test and training (left), and difference to reference accuracy obtained of Org data set as a train and test file.

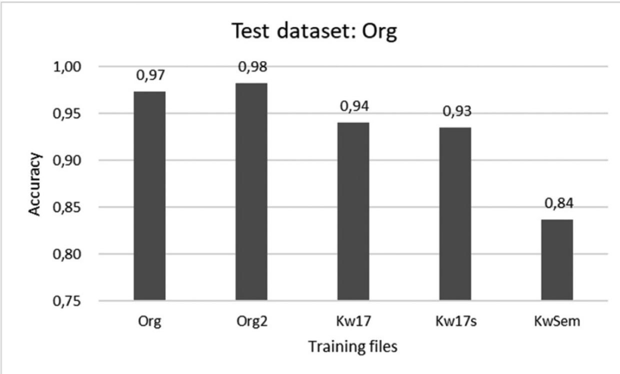

Figure 5. Testing with Org.

Table 7. Ambiguous category assigned to the same products.

Category	Product description
NIPPELE	15 x 1/2 conex nipple
MUFF	16 x 1/2 conex nipple, muff
CONNECTOR	3092 nipple knee bronze connector 1/2″
NIPPLE KNEE	3092 nipple knee bronze 3/4″
MODULE	AM Display Module 16A without sensor
SENSOR	AM Display Module 32A with sensor
ACTUATOR	Bimetallic Room Thermostat 24 V/230 V
THERMOSTAT	Bimetallic Room Thermostat 12 V/230 V
FILTER	Water slant valve filter PIZA PN25 1″
VALVE	Water slant valve filter PIZA PN25 1/2″

Table 8. Erroneous categorization and suggestions from classifiers.

Class predicted – class in file – an example
WIRE – predicted for CABLES: 5 examples, consider WIRE
CABLE – Power supply cable – type A
CABLE – Power supply cable – type B
CABLE – Control cable YGKY 22x0,8
CABLE – Solar cable 4 mm red
CABLE – Self-regulating heating cable, 5m
CONNECTOR predicted for TAPE: 2 examples, consider CONNECTOR
TAPE – Connector for colored LED tapes
TAPE – Quick mount connector for flat led tapes
FOIL predicted for TAPE: 3 examples, consider FOIL
TAPE – Yellow marker tape
TAPE – Universal marking tape
TAPE – Security tape marker

ambiguous UGC to product data incite a negative result of classification and missed search or browsing results. Thus, we can conclude that hypothesis H2 which states that *supplementing the professionally classified textual objects with UGC provide to ambiguous classification* is being verified positively.

Conclusions

The use of MLC algorithms in e-Commerce has become perceived as the common way for leveraging a company's competitive advantage.[36] An important component of online stores is centrally exposed search boxes. As a technology of search engines develops rapidly (it can be seen, e.g., from timetable of new releases of Elasticsearch[37] or Solr,[38]) the more application of MLC algorithms is in product management in order to support data quality for precise matching.[39] Assumption of our experiments was that UGC can positively influence training phase boosting classifiers to group products more friendly for users then based only on product names. Thus, we expected recall boosting keeping precision on good level. From user perspective, recall is perhaps even more important than precision because user can browse products in context of choice options, which increase the likelihood of purchase and satisfaction.[40] In this area, we have formulated hypothesis H1 which has been verified positively during investigation.

Effect of data ambiguity observed when we added UGC could be transferred to procedure of product data consistency testing when adding new products to web shop catalog. Erroneous categorization can be detected automatically, and suggestions of proper class can be considered based on confusion matrix, which is standard solution. This provides a great opportunity for application of automated procedure not only for product classification but also for feature management. During UDG inspection we have noticed that only 18,8% of search phrases do not include any digits. The question arises whether this is the language of communication in the industry or just a jargon of some individuals. Even assuming that UGC comprises mostly a subjective vocabulary, it is still a great opportunity to include some of them to the product data as supplementary description (selected synonymous) for bridging professional product names with jargon of user groups.

An important limitation of the research is a lack of clear relationship between proper classification as a helper system and a numeric evidence of improvement in search or browsing accuracy. Search engines can disguise some classification problems because they use ranking for product presentation, what makes that user can see several interesting products on the top of list, among others, what satisfies intent of the search, however, proper classification creates opportunity richer product presentation during typing (search-as-you-type)[41] and prompt of recommended options.

ORCID

Mieczysław Pawłowski http://orcid.org/0000-0002-9693-0573

References

1. Singh A, Tucker CS. A machine learning approach to product review disambiguation based on function, form, and behavior classification. Decis Support Syst. 2017;97:81–91. doi:10.1016/j.dss.2017.03.007.
2. Sun X, Han M, Feng J. Helpfulness of online reviews: examining review informativeness and classification thresholds by search products and experience products. Decis Support Syst. 2019;124:113099. doi:10.1016/j.dss.2019.113099.
3. Krumm J, Davies N, Narayanaswami C. User-generated content. IEEE Pervasive Comput. 2008;7(4):10–11. doi:10.1109/MPRV.2008.85.
4. Wang X, Li Y. How trust and need satisfaction motivate producing user-generated content. J Comput Inf Syst. 2017;57(1):49–57. doi:10.1080/08874417.2016.1181493.
5. Naab TK, Sehl A. Studies of user-generated content: a systematic review. Journalism. 2017;18(10):1256–73. doi:10.1177/1464884916673557.
6. Yi C, Jiang Z, Li X, Lu X. Leveraging user-generated content for product promotion: the effects of firm-highlighted reviews. Inf Syst Res. 2019;30 (3):711–25. doi:10.1287/isre.2018.0807.
7. Ukpabi DC, Karjaluoto H. What drives travelers' adoption of user-generated content? A literature review. Tour Manage Perspect. 2018;28:251–73. doi:10.1016/j.tmp.2018.03.006.
8. Lee S, Song J, Kim Y. An empirical comparison of four text mining methods. J Comput Inf Syst. 2010;51:1–10.
9. Kumar S, Kar AK, Ilavarasan PV. Applications of text mining in services management: a systematic literature review. Int J Inf Manage Data Insights. 2021;1 (1):100008. doi:10.1016/j.jjimei.2021.100008.
10. Li H, Liu H, Zhang Z. Online persuasion of review emotional intensity: a text mining analysis of restaurant reviews. Int J Hosp Manage. 2020;89:102558. doi:10.1016/j.ijhm.2020.102558.
11. Kowsari. K, Jafari Meimandi K, Heidarysafa. M, Mendu. S, Barnes. L, Brown. D. Text classification algorithms: a survey. Information. 2019;10(4):150. doi:10.3390/info10040150.
12. Mironczuk MM, Protasiewicz J. A recent overview of the state-of-the-art elements of text classification. Expert Syst Appl. 2018;106:36–54. doi:10.1016/j.eswa.2018.03.058.
13. Bernardi. L, Mavridis. T, Estevez P. 150 successful machine learning models: 6 lessons learned at booking.com. In Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining; 2019 Aug; Anchorage, AK. p. 1743–51.
14. Ingram J, Gaskell P. Searching for meaning: co-constructing ontologies with stakeholders for smarter search engines in agriculture. NJAS-Wageningen J Life Sci. 2019;90:100300. doi:10.1016/j.njas.2019.04.006.
15. Nelson P. Using engine scoring and predictive analytics to boost search accuracy. A Video Recap of Speaking Engagement at Lucene/Solr Revolution 2015; 2015.
16. Ma F, Mo Z, Luo Y. Empirical research on a model to measure end-user satisfaction with the quality of database search results. J Acad Librariansh. 2014;40 (2):194–201. doi:10.1016/j.acalib.2013.12.010.
17. Rokonuzzaman M, Harun A, Al-Emran M, Prybutok VR. An investigation into the link between consumer's product involvement and store loyalty: the roles of shopping value goals and information search as the mediating factors. J Retail Consum Serv. 2020;52:101933. doi:10.1016/j.jretconser.2019.101933.
18. Wakil K, Alyari F, Ghasvari M, Lesani Z, Rajabion L. A new model for assessing the role of customer behavior history, product classification, and prices on the success of the recommender systems in e-commerce. Kybernetes. 2019
19. Nigam P, Song Y, Mohan V, Lakshman V, Ding WA, Shingavi A, Yin B. Semantic product search. In Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining; 2019 Aug; Anchorage, AK. ACM. p. 2876–85.
20. Sandvig JC, Bajwa D. User perceptions of search enhancements in web search. J Comput Inf Syst. 2011;52:22–32.
21. Shvaiko P, Euzenat J. A survey of schema-based matching approaches. J Data Semant IV. 2005;146–71 in LNCS, volume 3730.
22. Jansen B, McNeese M. Evaluating the effectiveness of and patterns of interactions with automated searching assistance. J Am Soc Inf Sci Technol. 2005;56 (14):1480–503. doi:10.1002/asi.20242.
23. Koehn. D, Lessmann. S, Schaal. M. Predicting online shopping behaviour from clickstream data using deep learning. Expert Syst Appl. 2020;113342. doi:10.1016/j.eswa.2020.113342.
24. Gottlieb G, Lorimor TG (2017). U.S. Patent No. 9.646.095. Washington. DC: U.S. Patent and Trademark Office.
25. Kettunen E, Kemppainen T, Lievonen M, Makkonen M, Frank L, Kari T. Ideal types of online shoppers: a qualitative analysis of online shopping behavior. In Mediterranean Conference on Information Systems, 2018 Sep 28–30; MCIS; Corfu, Greece.
26. Hou T, Yannou B, Leroy Y, Poirson E. Mining customer product reviews for product development: a summarization process. Expert Syst Appl. 2019;132:141–50. doi:10.1016/j.eswa.2019.04.069.
27. Min H-J, Park JC. Identifying helpful reviews based on customer's mentions about experiences. Expert Syst Appl. 2012;39(15):11830–38. doi:10.1016/j.eswa.2012.01.116.
28. Nepomuceno MV, Laroche M, Richard MO. How to reduce perceived risk when buying online: the interactions between intangibility. product knowledge. brand familiarity. privacy and security concerns. J Retail Consum Serv. 2014;21(4):619–29. doi:10.1016/j.jretconser.2013.11.006.
29. Lee H, Yoon Y. Engineering doc2vec for automatic classification of product descriptions on O2O applications. Electron Commer Res. 2018;18(3):433–56. doi:10.1007/s10660-017-9268-5.
30. Zhang Y, Jin R, Zhou ZH. Understanding bag-of-words model: a statistical framework. Int J Mach Learn Cybern. 2010;1(1–4):43–52. doi:10.1007/s13042-010-0001-0.
31. Zhuo C. The Lao text classification method based on KNN. Procedia Comput Sci. 2020;166:523–38. doi:10.1016/j.procs.2020.02.053.
32. Tang Z, Li W, Li Y, Zhao W, Li S. Several alternative term weighting methods for text representation and classification. Knowl-Based Syst. 2020;207:106399. doi:10.1016/j.knosys.2020.106399.

33. Wu HC, Luk RWP, Wong KF, Kwok KL. Interpreting tf-idf term weights as making relevance decisions. ACM Trans Inf Syst. 2008;26(3):1–37. doi:10.1145/1361684.1361686.
34. Derczynski L. Complementarity. F-score. and NLP evaluation. In Proceedings of the Tenth International Conference on Language Resources and Evaluation (LREC'16); 2016 May; Portorož, Slovenia; p. 261–66.
35. Pedregosa F, Varoquaux G, Gramfort A, Michel V, Thirion B, Grisel O, … Vanderplas J. Scikit-learn: machine learning in Python. J Mach Learn Res. 2011;12:2825–30.
36. Bradlow ET, Gangwar. M, Kopalle. P, Voleti. S. The role of big data and predictive analytics in retailing. J Retail. 2017;93(1):79–95. doi:10.1016/j.jretai.2016.12.004.
37. https://www.elastic.co
38. https://solr.apache.org/
39. Hariharan. A, Gupta. A, Pal. T. CAMLPAD: cybersecurity autonomous machine learning platform for anomaly detection; 2019. arXiv preprint arXiv:1907.10442.
40. Belém FM, Silva RM, De Andrade CM, Person G, Mingote F, Ballet R, Gonçalves MA. "Fixing the curse of the bad product descriptions"–Search-boosted tag recommendation for E-commerce products. Inf Process Manage. 2020;57(5):102289. doi:10.1016/j.ipm.2020.102289.
41. Yin R, Li K, Zhang G, Lu J. A deeper graph neural network for recommender systems. Knowl-Based Syst. 2019;185:105020. doi:10.1016/j.knosys.2019.105020.